

Cardiovascular Disease in Persons With Chronic Kidney Disease

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CHAPTER OUTLINE

CARDIOVASCULAR DISEASE PATHOGENESIS IN CKD, 1203
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Cardiovascular disease is the leading cause of death across the spectrum of chronic kidney disease (CKD), with increased risk of cardiovascular disease seen in individuals with reduced glomerular filtration rate (GFR) and in those with even minimally elevated urine albumin excretion. The risk of cardiovascular disease outcomes increases as kidney function declines, with the risk of cardiovascular death in patients undergoing dialysis 10 to 20 times that of the general population. Cardiovascular disease has many manifestations in individuals with CKD, including stiff and noncompliant vessels and left ventricular hypertrophy (LVH); these lead to clinical syndromes such as ischemic heart disease, stroke, cognitive impairment, peripheral artery disease, heart failure, and arrhythmias.

In 1974, Lindner and colleagues¹ published a seminal article in the *New England Journal of Medicine*, describing the clinical courses of 39 patients with a mean age of 37 years at dialysis initiation who initiated maintenance hemodialysis in Seattle, Washington from 1960 to 1966. Over a mean follow-up of 6.5 years, 14 died of cardiovascular disease, with mortality rates more than 10-fold higher than in the general population. In describing these results, the authors coined the term “accelerated atherosclerosis,” perhaps a pathogenic misnomer but one that highlights the marked risk of cardiovascular disease in people with advanced CKD. Almost 25 years later, the National Kidney Foundation Task Force on Cardiovascular Disease in Chronic Renal Disease published a series of papers establishing the paradigm of the high burden of cardiovascular disease in CKD and outlining clinical strategies to reduce the high rates of cardiovascular morbidity and mortality in this population.² Since then, additional studies highlighted that increased cardiovascular disease risk can be seen at earlier stages of CKD,^{3,4} including in individuals with kidney damage but preserved GFR.⁵ This increased risk of cardiovascular disease likely represents accumulated risk from both traditional

risk factors that are associated with cardiovascular disease in the general population, such as diabetes, hypertension, and dyslipidemia, all of which are common in people with CKD, as well as nontraditional risk factors, such as mineral and bone disorder, inflammation, and anemia.⁶

The nature of cardiovascular disease in individuals with CKD is better understood now than at the time of the first report of the Seattle experience. While atherosclerosis with intimal plaque formation certainly contributes to arterial pathology, and subsequent plaque rupture or superficial erosion may lead to ischemic heart events,⁷ other nonatheromatous changes including arterial wall thickening and calcification appear to be the dominant pathologic features in persons with CKD contrasting with the general population,⁸ such that subacute to chronic ischemia, structural heart disease predisposing to heart failure, and cardiac arrhythmias disproportionately cause morbidity and mortality.⁹

This chapter concentrates on pathogenesis, risk factors, diagnosis, and treatment of subclinical and clinical cardiovascular disease in individuals with reduced GFR including those receiving dialysis, acknowledging that albuminuria, even in those with higher GFR levels, is a strong predictor of cardiovascular disease risk. The chapter focuses primarily on coronary artery disease, heart failure, and cardiac arrhythmias with less focus on other important types of cardiovascular disease including peripheral arterial disease and cerebrovascular disease.

CARDIOVASCULAR DISEASE PATHOGENESIS IN CKD

Manifesting with ischemia, heart failure, and arrhythmia, cardiovascular disease is overwhelmingly the leading cause

of morbidity and mortality in individuals with CKD. The risks of atherosclerotic cardiovascular disease events and heart failure are high in individuals with advanced CKD, consistent with the high prevalence of traditional cardiovascular disease risk factors like diabetes, hypertension, obesity, and dyslipidemia. However, the markedly higher overall risk of cardiovascular disease events likely represents a dramatic rise in the contribution of nonatherosclerotic cardiovascular events as kidney function declines (Fig. 42.1).¹⁰ In other words, at some point in the natural history of CKD, nonatherosclerotic cardiovascular disease including heart failure becomes the dominant pathology.

Cardiovascular findings can be divided broadly into those that involve the blood vessels (specifically the arteries) and those that involve the heart (Table 42.1). Although all of these findings are common in individuals with CKD, many appear disproportionately prevalent in individuals with advanced CKD, beyond that expected with traditional risk factors, such as older age, diabetes, and hypertension.

In most cases, clinically apparent cardiovascular disease reflects the interplay among these vascular and heart conditions. Both of these conditions may manifest

with ischemic heart disease and heart failure, and clinical disease often reflects the concurrent presence of both atherosclerotic disease and vascular remodeling. Certain risk factors including dyslipidemia primarily predispose an individual to development and progression of atherosclerosis, whereas others including mineral and bone disorder may predispose to arterial stiffness. Volume overload and anemia may primarily predispose an individual to cardiac remodeling and LVH. Hypertension, which is common at all stages of CKD, is associated with all of these disease manifestations. Over time, the interplay among these manifestations may yield both segmental myocardial perfusion defects due to disease affecting larger coronary arteries and insufficient subendocardial perfusion secondary to cardiac hypertrophy (causing increased demand) and capillary dropout. The end result is myocyte death and fibrosis, as well as increased risk of cardiac arrhythmias.

ARTERIAL DISEASE

The term “arteriosclerosis” (derived from the Greek meaning “hardening of the arteries”) is generally used to describe a range of pathologic processes impacting arteries. Historically, the term arteriosclerosis encompassed three different lesions: atherosclerosis, arteriolosclerosis, and Mönckeberg medial calcific sclerosis.¹¹ For the purposes of this chapter, we conceptualize these as **atherosclerosis** and **nonatheromatous arterial disease**.

Atherosclerosis is a disease of larger arteries characterized by an “atheroma,” with lesions enlarging the arterial intima that contain variable amounts and types of lipids, other cells, and extracellular components.¹² Intimal calcification is common in atherosclerosis, and its presence or absence is relevant in determining the stage of the lesion.¹² Simplistically, atherosclerosis can be considered an occlusive disease of the vasculature that occurs because of the deposition of lipid-laden plaques. Atherosclerosis is common in individuals with advanced CKD, in part reflecting the high burden of conditions predisposing to atherosclerosis including diabetes, dyslipidemia, hypertension, and other aspects of metabolic syndrome.

Nonatheromatous arterial disease is distinct from atherosclerosis and refers to the phenomenon of noncalcified, nonatheromatous stiffening of smaller muscular arteries. In 1903, Mönckeberg reported an additional form of nonatheromatous arterial disease involving the media of the artery and characterized by medial thickening and heavy calcification.¹³ Nonatherosclerotic disease is better characterized on the basis of its effect, which is typically thickening of the vessel and a loss of arterial elasticity (stiffening); these may reflect primary arterial calcification, fibromuscular intimal thickening, and/or arterial hyalinosis.¹⁴ Accordingly, arterial stiffness reflects nonocclusive remodeling of the vasculature accompanied by a loss of arterial elasticity.

Critically, individuals with advanced CKD may exhibit all of the above features of arterial disease. Endothelial dysfunction, whereby the vasculature is unable to constrict and dilate to match supply to demand, further contributes to clinical manifestations of structural arterial disease. The nonatheromatous structural and functional arterial

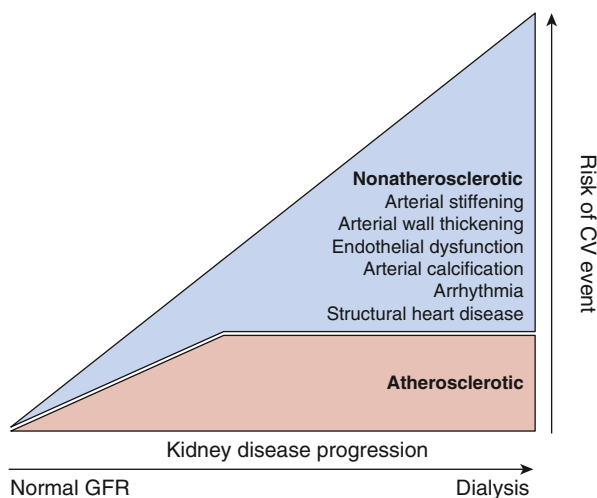


Fig. 42.1 Depiction of the relative contributions of cardiovascular disease types across chronic kidney disease stages. Atherosclerotic and nonatherosclerotic disease may contribute synergistically to increased cardiovascular risk as kidney function declines. CV, Cardiovascular; GFR, glomerular filtration rate. (Adapted from Wanner C, Amann K, Shoji T. The heart and vascular system in dialysis. *Lancet*. 2016;388:276–284.)

Table 42.1 Manifestations of Cardiovascular Diseases in Persons With Chronic Kidney Disease

Arteries	Heart
Increased wall thickness	Altered cardiac geometry
Arterial stiffness	Myocardial fibrosis
Endothelial dysfunction	Left ventricular dysfunction
Arterial calcification	Valvular disease
Atherosclerosis	Dysrhythmia and conduction defects

abnormalities most associated with CKD are covered in more detail later.

ARTERIAL WALL THICKENING

One of the few autopsy studies examining arterial pathology in patients receiving dialysis reported that there was more pronounced medial thickening in sections of coronary arteries than in equivalent sections from age- and sex-matched patients without CKD known to have had coronary artery disease.⁸ Thickening of the arterial wall can be measured noninvasively by assessing the combined width of the intima and media of the carotid artery on ultrasound (carotid intima-media thickening; CIMT) and associates with high cardiovascular disease risk.^{15,16} Increased CIMT is also found in individuals with less advanced CKD,¹⁷ with arterial wall thickening likely reflecting the kidney disease milieu rather than other comorbid conditions as suggested from data from children with kidney disease.¹⁸

ARTERIAL STIFFENING

Arterial stiffening is likely a functional consequence of artery wall thickening and calcification and is readily assessed noninvasively by measuring the velocity of propagation of a pulse wave through the arterial tree (pulse wave velocity, PWV).¹⁹ Stiffening may represent an early feature of CKD-associated arterial disease and can be detected in children receiving dialysis as early as the first decade of life.²⁰ Arterial stiffness has prognostic significance: Both carotid²¹ and aortic²² stiffness independently predict death in individuals with CKD including those receiving hemodialysis.²³ Several additional methods are available to assess arterial stiffness including the carotid augmentation index, which provides a measure of the interaction between outgoing and reflected pulse waveforms at the point of measurement and, in part, reflects the stiffness of the arterial tree²⁴ and carotid artery stiffness, which uses ultrasound to measure pulsatile changes in the artery.²⁵

ENDOTHELIAL DYSFUNCTION

The vascular endothelium plays a key role in maintaining arterial tone, predominantly through the continuous production of nitric oxide. Nitric oxide is a vasoactive compound that contributes to the resting tone of the artery and protects against the development of arterial disease by inhibiting vascular smooth muscle cell proliferation, platelet aggregation, and monocyte adhesion.²⁶ Production of nitric oxide is stimulated by hypoxia, increased shear stress, or locally released mediators such as acetylcholine. Several methods are used to measure endothelial function including flow-mediated dilation for larger, peripheral vessels and laser Doppler flowmetry for the microvasculature, as well as more invasive techniques such as assessing the response to intracoronary infusion of acetylcholine.²⁷ Most of these are limited to research uses and have issues with intraindividual and interindividual reproducibility.

Endothelial function is often impaired in individuals with advanced CKD.^{28,29} Asymmetric dimethyl arginine (ADMA) is associated with impaired endothelial function in CKD.³⁰ ADMA inhibits nitric oxide synthetase, thereby limiting the bioavailability of nitric oxide, which is essential for normal endothelial function.³¹ Blood concentrations of ADMA are elevated in CKD, are inversely proportional to

GFR,^{30,32} and appear to be associated with an increased risk of cardiovascular disease and mortality in studies in both the general population and in individuals with CKD.³³⁻³⁵ At present there is no intervention that can selectively reduce ADMA concentrations; accordingly, the causal relationship (and clinical relevance) of this risk factor remains unclear. The major consequence of endothelial dysfunction is a supply-demand mismatch, such that the vasculature is unable to compensate to meet metabolic demands, resulting in ischemia. This may be a major mechanism in the development and progression of end-organ disease including heart disease in people with CKD.

ARTERIAL CALCIFICATION

Arterial calcification is a frequent feature of arterial disease in individuals with CKD.^{13,36} There are two broad phenotypes: intimal calcification and medial calcification. Intimal and medial calcification appear similar on imaging, although medial calcification, for unknown reasons, is far less common in vessels that are prone to atherosclerosis, such as the coronary arteries.³⁷

Intimal calcification, which is often patchy and associated with lipid deposits, is a frequent finding in atherosclerosis. Intimal calcification typically affects medium and large arteries and, as discussed earlier, is common in both the general population and people with CKD, reflecting its relationship with traditional cardiovascular disease risk factors like diabetes, hypertension, and dyslipidemia.

Medial calcification is more specific to CKD,³⁸ often manifesting with a linear pattern of medial calcium deposition (Fig. 42.2).³⁹ Medial calcification may be more closely associated with abnormalities in mineral and bone metabolism that are common in advanced CKD, although there is considerable heterogeneity in the prevalence of medial calcification, even within a hemodialysis population. Calcific uremic arteriolopathy, also dubbed calciphylaxis, is an aggressive form of calcification that is often coexistent with medial calcification and is discussed in detail in [Chapter 52](#).

Mechanisms of arterial calcification remain uncertain but likely represent an imbalance between calcification promoters and calcification inhibitors occurring within the uremic milieu and are further exacerbated by a phenotypic change in vascular smooth muscle cells toward an osteochondrogenic phenotype.³⁶ Calcification inhibitors include vitamin K-dependent carboxylated matrix Gla protein, fetuin-A, pyrophosphate, and osteopontin, with FGF-23-dependent Klotho potentially also implicated. Lower levels of these inhibitors predispose to arterial calcification. Similarly, elevated levels of calcification promoters also predispose to arterial calcification; these include alkaline phosphatase, calprotectin, osteocalcin, and inflammatory cytokines, with inflammation potentially explaining some of the heterogeneity in the prevalence of medial calcification.⁴⁰ Calcification likely includes passive processes, whereby calcium containing compounds precipitate and are deposited into the vasculature, as well as active processes, most notably via the phenotypic changes in vascular smooth muscle cells noted earlier.⁴¹

To date, there are no specific treatments to prevent arterial calcification, with many clinical trials unable to show substantial clinical benefits when targeting a single

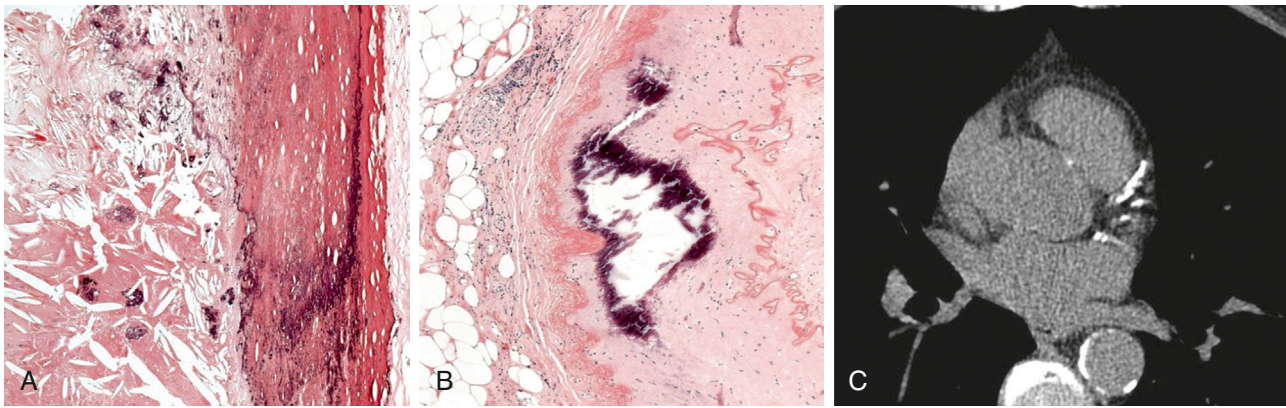


Fig. 42.2 Arterial calcification in chronic kidney disease (CKD). Cross-sections of medium-sized arteries from a patient with CKD showing deposition of calcium (black) in the intima (A) and media (B) in association with atherosclerosis (von Kossa stain). Calcium deposits may be visible on computed tomography scanning of the heart as depicted in (C) where calcification is visible in the left anterior descending and left circumflex coronary artery, as well as the descending aorta. (Courtesy Professor A.J. Howie.)

contributor to this phenomenon.⁴²⁻⁴⁶ That stated, dietary phosphorus control is likely a reasonable intervention and, among individuals with a propensity to arterial calcification, avoidance of the vitamin K inhibitor, warfarin, may also be reasonable.

STRUCTURAL HEART DISEASE

Structural heart disease is common in people with CKD and includes LVH, myocardial fibrosis, and valve disease. Although often adaptive in the early stages, these structural changes may eventually lead to functional impairment including reduced compliance of the left ventricular wall during diastole, a classic feature of heart failure with preserved ejection fraction (HFpEF), and impaired myocardial contractility, as is seen in heart failure with reduced ejection fraction (HFrEF).⁴⁷ Critically, both reduced LV compliance and impaired myocardial contractility may occur concurrently. In addition, histologic changes such as fibrosis and calcification occur in the myocardium,⁴⁸ and valvular calcifications are frequently observed.⁴⁹ This section focuses on structural heart disease itself, while clinical syndromes that are associated with structural heart disease, such as heart failure, are discussed later in this chapter.

LEFT VENTRICULAR HYPERTROPHY

LVH affects up to 75% of patients initiating hemodialysis and is highly prevalent among patients with earlier stages of CKD.^{50,51} The diagnosis of LVH is readily made with echocardiography, an inexpensive, noninvasive, and widely available test. Cardiac function should be assessed in the euvoletic state, as both significant volume depletion and overload may reduce left ventricular inotropy. Accordingly, in patients undergoing dialysis, two-dimensional echocardiography is likely to be most informative if performed on the interdialytic day and, notably, is less vulnerable to volume fluctuation than 1D (M-mode) echocardiography methods, which are seldom used today. Other imaging techniques including more advanced echocardiography methods, cardiac magnetic resonance imaging (MRI), and cardiac positron emission tomography may have additional

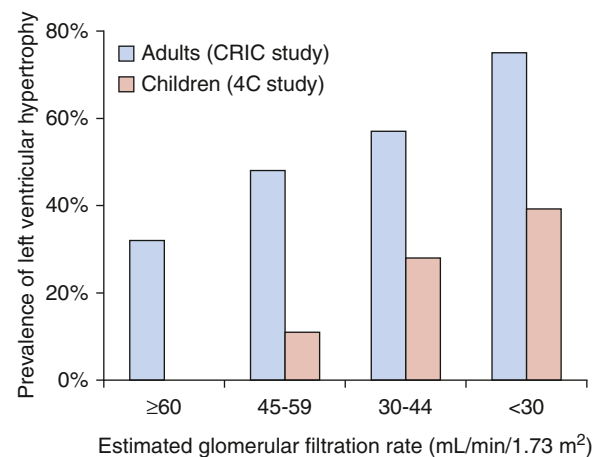


Fig. 42.3 Prevalence of left ventricular hypertrophy in adults and children with chronic kidney disease.

utility, including assessment of cardiac fibrosis, albeit at greater cost and, for MRI, the use of gadolinium. Screening echocardiography has been recommended for incident patients undergoing dialysis; however, there is no evidence that routinely obtaining echocardiograms improves clinical outcomes.

LVH prevalence is higher at lower GFR levels. In the Chronic Renal Insufficiency Cohort (CRIC) study of 3487 patients with CKD, the prevalence of LVH was 32%, 48%, 57%, and 75% among individuals with estimated GFR ≥ 60 , 45 to 59, 30 to 44, and < 30 mL/min/1.73 m², respectively (Fig. 42.3).⁵² These findings contrast with a prevalence of LVH of $< 20\%$ in older adults in the general population. Highlighting the association between CKD and LVH is the GFR-dependent increase in LVH prevalence also seen among children with CKD, in whom other confounding conditions are less common (see Fig. 42.3).⁵³

LVH may manifest with left ventricular wall thickening alone, referred to as concentric hypertrophy, or with increased cavity volume accompanied by wall thickening, referred to as eccentric hypertrophy. Chronic volume overload leads to increased left ventricular filling pressure,



Fig. 42.4 Left ventricular disease in chronic kidney disease (CKD). A cross-section of a postmortem heart from a patient with longstanding CKD showing concentric left ventricular hypertrophy (A). Histologic analysis (B) often reveals myocardial fibrosis (pale staining) disrupting the normal architecture of cardiac myocytes. (Courtesy Professor A.J. Howie [A] and Dr. M. Rubens [B].)

stretching the LV wall. The heart adapts by lengthening existing myocytes, thereby enlarging the internal dimensions of the LV. This process is usually accompanied by wall thickening, a further adaptive response that reduces wall stress. The end result of chronic volume overload is a ventricle with a thickened wall and enlarged cavity but with a normal wall thickness to internal diameter ratio (eccentric hypertrophy). In contrast, pressure overload, as seen with hypertension and aortic stenosis, increases wall stress during systole, leading to myocyte proliferation and wall thickening with either preservation or reduction of cavity volume (concentric hypertrophy; see Fig. 42.4). These responses are initially adaptive: dilatation permits increased cardiac output for a similar level of energy expenditure, while wall thickening redistributes increased tension over a larger area and reduces energy consumption per myocyte.⁵⁴ However, over time, particularly in conjunction with progressive arterial disease or valvular disease, they may become maladaptive, leading to HFpEF and increased morbidity and mortality.⁵⁵

MYOCARDIAL FIBROSIS

A mismatch between demand and myocyte oxygen supply over time leads to cell death and interstitial cardiac fibrosis (Fig. 42.5).⁴⁸ This is a multifactorial process, with increased myocyte work increasing oxygen demand while ischemia, impaired angiogenesis, and impaired endothelial function, particularly in the setting of stiffened conduit arteries, result in decreased perfusion. Even in the absence of occlusive coronary artery lesions, there often is a reduction in capillary density to hypertrophied cardiac myocytes, exacerbating local hypoxia.⁵⁶ Furthermore, stiffening of conduit arteries leads to a fall in diastolic pressure, which in turn may compromise coronary artery perfusion during diastole. Finally, it is possible that repetitive myocardial ischemia

induced by hemodialysis exacerbates myocyte injury.⁵⁷ This chronic mismatch in oxygen supply and demand to the myocardium might explain the well-recognized clinical observation that patients receiving dialysis experience chest pain,⁵⁸ even in the absence of occlusive lesions in the major epicardial coronary arteries (i.e., “demand ischemia”) and have chronically elevated levels of cardiac damage biomarkers.⁵⁹

VALVE DISEASE

Other structural heart diseases seen commonly in CKD include aortic valve calcification and stenosis, mitral valve and mitral annular calcification, and mitral and tricuspid regurgitation.^{60,61} Mitral valve or mitral annular calcification was present in 20% of individuals with reduced kidney function (roughly stage 3–4 CKD) in the Framingham Offspring Study.⁶² Similarly, in the CRIC study, lower estimated GFR was strongly associated with increased likelihood of mitral annular calcification, with risk increased by 50%, 130%, 226%, and 278% for estimated GFR categories 50 to 60, 40 to 50, 30 to 40, and <30 mL/min per 1.73 m², respectively, when compared with estimated GFR ≥60 mL/min per 1.73 m².⁶³ The Framingham Heart Study and other studies also show a high prevalence of aortic valve calcification in individuals with CKD,⁶² with studies in CRIC demonstrating a “dose-dependent” association between lower estimated GFR and greater aortic valve calcification that was independent of traditional cardiovascular risk factors.⁶⁴

Patients undergoing hemodialysis also have a high prevalence of valvular calcification; in one review, the prevalence of mitral valve calcification ranged from 25% to 59% and aortic valve calcification from 28% to 55%; this compares with expected prevalence of 3% to 5% in the general population.⁶⁵ Studies have demonstrated rates of

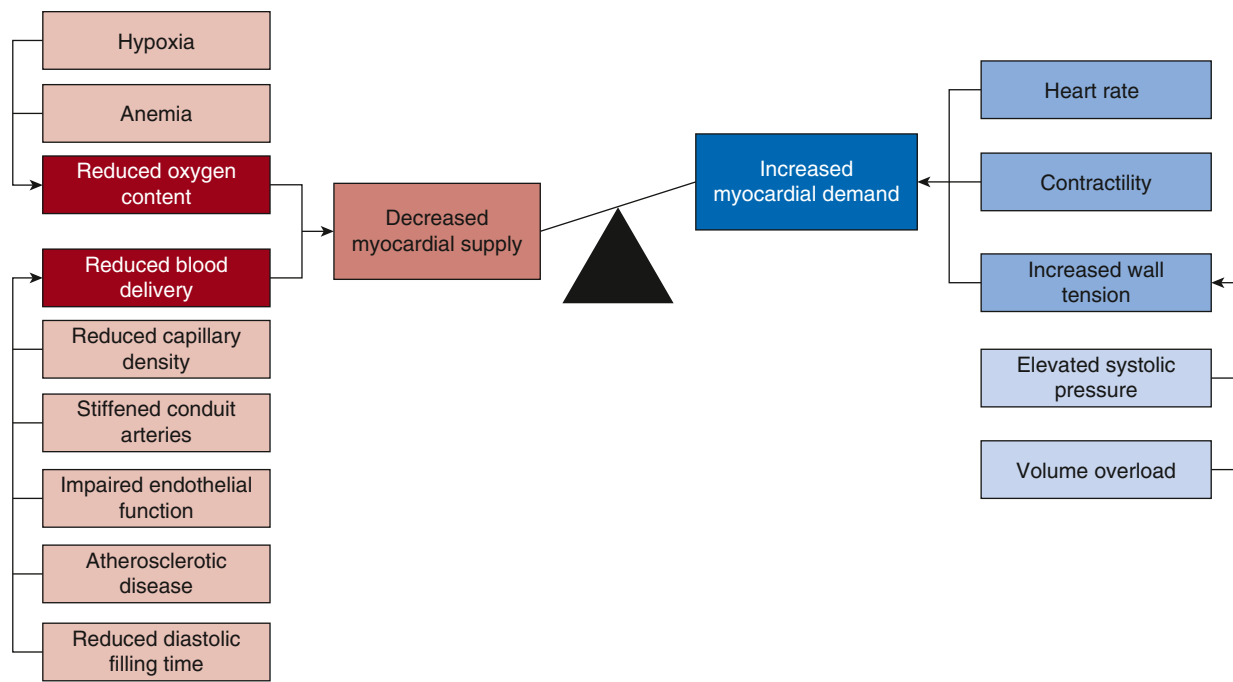


Fig. 42.5 Pathogenesis of myocardial fibrosis as a function of oxygen supply-demand mismatch. Multiple factors result in decreased supply (reduced oxygen content and blood delivery) and increased demand (increased heart rate, contractility, and wall tension).

mitral annular calcification ranging from 30% to 50% in patients undergoing hemodialysis.⁶⁵ The implications of valve disease are discussed later in this chapter in the section on heart failure.

RISK FACTORS FOR CARDIOVASCULAR DISEASE

Much of the increased burden of cardiovascular disease in CKD is a result of increased prevalence of both traditional and nontraditional cardiovascular disease risk factors. Traditional risk factors were identified in the Framingham Heart Study as conferring increased risk of cardiovascular disease in the general population (Table 42.2). Nontraditional risk factors were not defined in the initial reports of the Framingham Heart Study but increase in prevalence as kidney function declines and are hypothesized to be cardiovascular disease risk factors in individuals with CKD (see Table 42.2). Though CKD, particularly late-stage CKD, may directly cause cardiovascular disease through mechanisms that include fluid retention, anemia, abnormal mineral metabolism, and inflammation, it is likely that CKD also represents a risk state in which greater severity of factors associated with the development of CKD (including diabetes, hypertension, and possibly dyslipidemia) also contribute to enhanced cardiac risk. In the latter hypothesis, the presence of CKD may be a marker of the severity and duration of these other risk factors. Traditional risk factors including diabetes and hypertension are addressed elsewhere in this book (see Chapters 41 and 46). The remainder of this section reviews dyslipidemia and key nontraditional risk factors.

Table 42.2 Traditional and Nontraditional Cardiac Risk Factors in Chronic Kidney Disease

Traditional Risk Factors	Nontraditional Factors
Older age	Albuminuria
Male sex	Lipoprotein (a) and apo (a) isoforms
Hypertension	Lipoprotein remnants
Higher LDL cholesterol	Anemia
Lower HDL cholesterol	Abnormal mineral metabolism
Diabetes	Extracellular fluid volume overload
Smoking	Electrolyte abnormalities
Physical inactivity	Oxidative stress
Menopause	Inflammation
Family history of cardiovascular disease	Malnutrition
Left ventricular hypertrophy	Thrombogenic factors
	Sleep disturbances
	Altered nitric oxide/endothelin balance
	Sympathetic overactivity

HDL, High-density lipoprotein; LDL, low-density lipoprotein.

From Sarnak MJ, Levey AS. Cardiovascular disease and chronic renal disease: a new paradigm. *Am J Kidney Dis.* 2000;35(4 suppl 1):S117–S131.

DYSLIPIDEMIA

The characteristic lipid profile in individuals with CKD, specifically those with low GFR and those with severely elevated albuminuria, is an accumulation of partially catabolized triglyceride-rich very-low-density lipoprotein and

intermediate-density lipoprotein (IDL) particles, leading to elevated serum triglyceride concentrations, with lower high-density lipoprotein (HDL) cholesterol concentrations. Low-density lipoprotein (LDL) cholesterol concentration in CKD is similar or lower than the population average, except that nephrotic-range proteinuria leads to an increase in LDL cholesterol.⁶⁶ Cholesterol continues to be progressively removed from these particles and triacylglycerol is added, leading to an excess of small, dense LDL particles that may be more atherogenic.⁶⁷ HDL function may also be impaired in CKD.⁶⁸ In addition, lipoprotein (a) (Lp[a]) concentrations are higher in advanced CKD.⁶⁹ Critically, as discussed later in “Ischemic Heart Disease and Cardiovascular Death,” treatments targeting dyslipidemia in individuals with advanced CKD, particularly those receiving dialysis, have been less impactful for reducing adverse cardiovascular outcomes than similar interventions in the general population.

ANEMIA

Anemia in CKD reflects the interplay among relative or absolute erythropoietin deficiency, functional iron deficiency, and chronic inflammation (see [Chapter 53](#)). In individuals with CKD, anemia is associated with LVH: In one study, a 0.5 g/dL lower hemoglobin concentration was associated with a 30% higher frequency of increased left ventricular mass (defined as 20% greater than baseline),⁷⁰ while in a study of patients receiving dialysis from the 1990s, each 1 g/dL lower hemoglobin was associated with a 50% increased risk of left ventricular dilatation and a 25% increased risk of cardiac failure.⁷¹

Although nonrandomized studies suggested that partial correction of anemia reduced left ventricular mass index in patients with CKD,^{72,73} early, small randomized trials did not show that targeting normalization of hemoglobin leads to a reduction in left ventricular mass.⁷⁴⁻⁷⁶ Reinforcing this finding, the large clinical trials that evaluated using epoetin or darbepoetin to target hemoglobin normalization in individuals with advanced CKD showed no cardiovascular benefits and potential suggestions of harm.⁷⁷⁻⁸⁰ Viewed in sum, these results suggest that anemia is associated with cardiovascular disease but that currently used therapies to treat anemia do not yield significant cardiovascular benefits.

CHRONIC KIDNEY DISEASE MINERAL AND BONE DISORDER

Individuals with advanced kidney disease experience a complex disorder of calcium and phosphate metabolism including effects on vitamin D, parathyroid hormone, and fibroblast growth factor (FGF)-23 (see [Chapter 52](#) for a detailed description). While, historically, CKD-mineral and bone disorder (MBD) has been focused on bone health, there is increasing recognition that CKD-MBD has considerable impacts on vascular health, most likely by predisposing to arterial calcification and stiffening, with vessels developing an osteoblastic phenotype. While specific interventions targeting CKD-MBD for cardiovascular disease prevention and treatment have been disappointing, potentially representing the complex interrelationship among CKD-MBD mediators and cardiovascular disease, ongoing research continues in this space.

For example, while hyperphosphatemia is associated with a higher risk of cardiovascular disease events, trials have not

consistently shown a benefit of phosphate-lowering therapies on cardiovascular disease reduction.^{44,81} This may reflect the population selected or the interventions used. Similarly, one large trial evaluating the effects of parathyroid hormone lowering with cinacalcet on cardiovascular disease events found that, in an unadjusted intention-to-treat analysis, cinacalcet did not significantly reduce the risk of death or major cardiovascular events in patients with moderate-to-severe secondary hyperparathyroidism who were undergoing dialysis, although post-hoc analyses adjusting for baseline characteristics demonstrated nominally significant benefits, as did analyses censoring data for patients after kidney transplantation, parathyroidectomy, or use of commercially available cinacalcet.⁴² Additionally, a meta-analysis of vitamin D therapies in adults with CKD did not show definitive benefits on cardiovascular disease outcomes.⁸²

FGF-23 concentrations rise earlier in CKD than the serum concentrations of phosphate or even parathyroid hormone⁸³ and help maintain a normal serum phosphate concentration via a phosphaturic effect. The association of FGF-23 with mortality was first demonstrated among patients receiving maintenance hemodialysis,⁸⁴ with subsequent studies demonstrating a robust association of between higher FGF-23 concentrations and all-cause and cardiovascular mortality in patients with CKD,⁸⁵⁻⁸⁷ potentially driven by heart failure events.⁸⁸ Experimentally, FGF-23 has been shown to induce LVH in the absence of its coreceptor klotho,⁸⁹ and FGF-23 levels are associated with LVH in both adults and children with CKD.^{90,91} Whether FGF-23 is causal of cardiovascular outcomes in the clinical setting remains uncertain,⁹² and more data are needed on therapies to lower FGF-23, as well as the potential cardiovascular effects of FGF-23 lowering in patients with CKD.

OXIDATIVE STRESS AND INFLAMMATION

Oxidative stress and inflammation are associated with cardiovascular disease and are more common in individuals with CKD. While it would follow that interventions to reduce inflammation and oxidative stress, including use of antioxidants, could favorably impact cardiovascular disease, results have been disappointing to date. A 2023 Cochrane meta-analysis of 95 clinical trials assessing vitamin and nonvitamin antioxidants in individuals with CKD found variable results for interventions, highlighting small sample sizes and low trial quality.⁹³ Currently there is no role for routine use of these agents for cardiovascular disease prevention in people with CKD, although further trials are ongoing.

DIAGNOSIS, PREVENTION, AND TREATMENT OF CARDIOVASCULAR DISEASE ASSOCIATED WITH CHRONIC KIDNEY DISEASE

ISCHEMIC HEART DISEASE AND CARDIOVASCULAR DEATH

EPIDEMIOLOGY

Chronic kidney disease stages 3–4

The prevalence of cardiovascular disease is high at all stages of CKD, rising as kidney function declines. On the basis

of age, sex, and race-adjusted U.S. Medicare data, mostly composed of older adults, CVD was present in 37.5% of individuals without CKD compared with 63% of patients with diagnostic codes for CKD stages 1 to 2 CKD, 67% of patients with CKD stage 3, and 75% of patients with CKD stages 4 to 5 (Fig. 42.6).⁹⁴ Critically, both reduced estimated GFR and moderately or severely increased albuminuria with preserved estimated GFR (urine albumin-to-creatinine ratio 30 to 299 mg/g or ≥ 300 mg/g, indicating CKD stages 1 to 2) are independently associated with prevalent cardiovascular disease.

There is also a progressive increase in the age-standardized incidence of cardiovascular disease events as kidney function declines, such that, compared with 21 cardiovascular events per 1000 person-years among individuals with estimated GFR >60 mL/min per 1.73 m², rates increase to 37, 113, 218, and 366 events per 1000 person-years among people with CKD stage 3a, CKD stage 3b, CKD stage 4, and CKD stage 5, respectively.⁹⁴ Even in adjusted analyses, the risk of cardiovascular death is dramatically increased at lower GFR (Fig. 42.7).⁹⁵ The risk relationship between estimated GFR and cardiovascular disease events is independent of a person having preexisting cardiovascular disease.⁹⁵

Similar associations are seen in individuals with functioning kidney transplants. Among kidney transplant recipients, each 5 mL/min/ 1.73 m² higher estimated GFR below 45 mL/min/ 1.73 m² is independently associated with a 15% lower risk of CVD.⁹⁶ Similarly, the presence of albuminuria, independent of the estimated GFR, is also associated with a higher risk of cardiovascular disease events among kidney transplant recipients.⁹⁷

Chronic kidney disease stage 5/dialysis

In patients undergoing dialysis, both prevalent and incident cardiovascular diseases are common. For

example, a 30-year-old patient undergoing dialysis has similar cardiovascular mortality as an 80-year-old individual from the general population (Fig. 42.8). This likely reflects a high prevalence of underlying cardiovascular disease (~43% of prevalent patients in the United States in 2018 had a diagnosis of coronary artery disease and ~43% had a congestive heart failure diagnosis), as well as a higher case-fatality rate than the general population.⁹⁸

DIAGNOSIS OF ISCHEMIC HEART DISEASE

Ischemic heart disease may be acute or chronic. In the acute setting, diagnosis hinges on a combination of biomarkers, electrocardiogram results, imaging tests, and clinical scenarios, while imaging is most widely used for identifying chronic ischemia. Diagnosis of an acute myocardial infarction requires clinical evidence of acute myocardial ischemia and with detection of a rise and/or fall of cardiac troponin values with at least one value above the 99th percentile upper reference limit and at least one of the following: symptoms of myocardial ischemia; new ischemic ECG changes; development of pathologic Q waves; imaging evidence of new loss of viable myocardium or new regional wall motion abnormality in a pattern consistent with an ischemic etiology; or identification of a coronary thrombus by angiography or autopsy.⁹⁹ Studies have suggested that diagnosis of acute myocardial infarction may be challenging in patients with advanced stages of CKD, in part due to the presence of chronically elevated myocardial injury biomarkers.⁵⁸ Patients treated with kidney disease may present with more severe manifestations of cardiac ischemia.¹⁰⁰

There are two major subtypes of acute myocardial infarction (MI): type 1 MI, which is caused by atherothrombotic coronary artery disease and most often

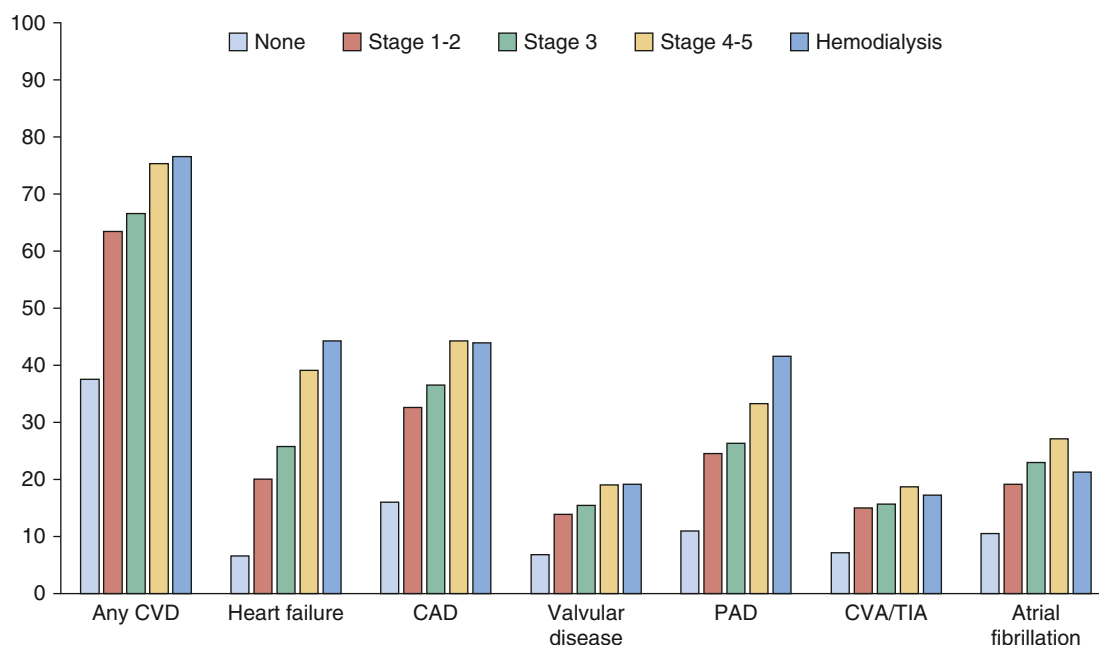


Fig. 42.6 Age-, sex-, and race-adjusted prevalence of cardiovascular disease (CVD) and its subtypes in individuals with chronic kidney disease. CAD, Coronary artery disease; CVA, cerebrovascular accident; PAD, peripheral artery disease; TIA, transient ischemic attack. (Data are derived from the USRDS 2020 Annual Data Report, Volume 1, Chapter 4 and Volume 2, Chapter 8. Available at: <https://adr.usrds.org/2020/chronic-kidney-disease/4-cardiovascular-disease-in-patients-with-ckd>. Accessed September 29, 2023.)

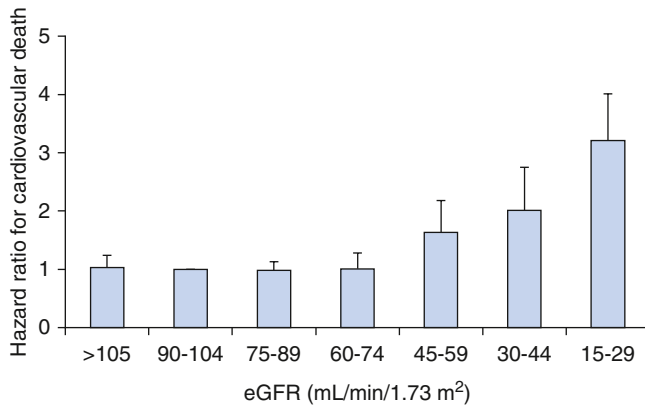


Fig. 42.7 Hazard ratios for cardiovascular events according to the baseline estimated glomerular filtration rate. Adjusted for age, sex, race, cardiovascular disease history, smoking status, diabetes mellitus, systolic blood pressure, serum total cholesterol, and urine albumin-to-creatinine ratio. (Plotted with data from van der Velde M, Matsushita K, Coresh J, et al. Lower estimated glomerular filtration rate and higher albuminuria are associated with all-cause and cardiovascular mortality. A collaborative meta-analysis of high-risk population cohorts. *Kidney Int.* 2011;79:1341–1352.)

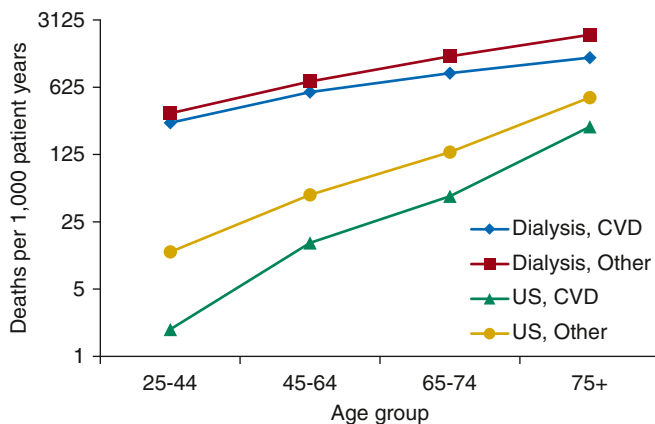


Fig. 42.8 Cardiovascular disease (CVD) and noncardiovascular disease mortality in the general U.S. population (2014) compared with patients with chronic kidney failure treated by dialysis (2012–2014). Cardiovascular disease (CVD) deaths in the U.S. population include “Diseases of the heart” (I00–I09, I11, I13, I20–I51) and “Cerebrovascular diseases” (I60–I69). CVD death in dialysis includes myocardial infarction, pericarditis, atherosclerotic heart disease, cardiomyopathy, arrhythmia, cardiac arrest, valvular heart disease, congestive heart failure, and cerebrovascular disease. The youngest general population group is 22 to 44 years old. (Reprinted from Gilbert SJ, Weiner DE, Bombardieri AS, et al. *The Primer on Kidney Diseases*. 8th ed. Philadelphia: Elsevier; 2022.)

precipitated by atherosclerotic plaque disruption; and type 2 MI, which is caused by a mismatch between cardiac oxygen supply and demand that is not in the setting of acute atherosclerotic plaque disruption.⁹⁹ Further, a type 3 MI is defined as an event resulting in cardiac death in the setting of symptoms suggestive of myocardial ischemia accompanied by new ischemic electrocardiogram changes, but where death occurs before biomarkers are assessed. This may be common among individuals with advanced CKD, where sudden cardiac death is a leading cause of mortality.

Imaging tests

No one diagnostic test is optimal for identifying ischemic heart disease in patients with CKD, and each has pitfalls specific to CKD that may affect sensitivity and specificity (Table 42.3).¹⁰¹ Currently, a functional assessment of perfusion that includes cardiac imaging is likely the best initial option to identify cardiac ischemia. Widely available options include exercise or pharmacologic nuclear stress tests, as well as exercise or pharmacologic stress echocardiography.¹⁰² Importantly, the ability to perform exercise stress testing is often limited by comorbid conditions in the CKD population. Dobutamine stress echocardiography, assuming adequate institutional expertise, may be a good option for detecting angiographically apparent coronary lesions while additionally providing information on valvular and other structural diseases.¹⁰³

Coronary artery calcium assessment and coronary computed tomography angiography have become increasingly popular for noninvasive assessment of cardiovascular risk in the general population. Coronary artery calcium score may be able to offer prognostic information, as higher scores are associated with worse outcomes in all CKD stages; however, its role in guiding treatment decisions in the advanced CKD population remains uncertain.¹⁰¹ Cardiac MRI, particularly in conjunction with pharmacologic stress, can provide extensive information, although imaging requires gadolinium. While concerns regarding gadolinium have diminished in individuals with CKD,¹⁰⁴ given some uncertainty regarding safety, particularly in patients receiving dialysis, other imaging modalities may be preferable. Cardiac positron emission tomography is becoming more widely available and may be useful to identify areas of ischemia and coronary flow reserve.¹⁰⁵ Left heart catheterization is typically considered the gold standard assessment of coronary artery disease, providing both direct assessment of flow-limiting lesions and a concurrent opportunity for intervention. It has little role as an initial screening test given its invasive nature.

Despite concerns regarding iodinated contrast in individuals with CKD, there is no absolute contraindication to its administration including for individuals receiving dialysis,¹⁰⁶ although preservation of existing kidney function is an important consideration in all stages of kidney disease, especially those treated with peritoneal dialysis. Given this, the risks and benefits associated with contrast-based diagnostic tools must be carefully assessed, although with careful management and conservative use of iodinated contrast, many individuals with advanced CKD can safely receive iodinated contrast (see Chapter 24 for a more detailed discussion of the use of gadolinium and iodinated contrast in people with CKD).

Biomarkers

Troponin is the gold standard biochemical marker for acute myocardial injury and, particularly in the advanced CKD/dialysis population, may be chronically elevated. Both cardiac troponin I and troponin T are widely available, with most assays reporting the results of high-sensitivity testing. These high-sensitivity assays can detect even mild ischemia.

Importantly, elevated troponin levels, above the 99th percentile for the assay, are fairly common among individuals with CKD screened in the absence of acute

Table 42.3 Imaging Modalities for Coronary Disease

	Stress Echocardiography	Gated SPECT Myocardial Perfusion Imaging	Gated PET Myocardial Perfusion Imaging	Coronary CT Angiography	Pharmacologic Stress Cardiac MR	Left Heart Catheterization
Strengths	-Assesses heart structure and function -Identifies regional wall abnormalities -Good sensitivity and superior specificity to MPI	-Widely available -Good sensitivity and specificity -Provides information on LV function	-Highly accurate -Provides information on myocardial blood flow	-Aligns with prognosis	-Combined assessment of LV function, myocardial perfusion, scarring and wall motion -Does not require the geometric assumptions used in 2D echo	-Allows for concurrent diagnosis and evaluation -Assessment of flow limiting lesions
Limitations	-Poor acoustic windows may limit accuracy -Inadequate stress may result in a false negative -User expertise	Soft tissue attenuation and artifacts may decrease test performance	-Expensive -Limited availability	-Limited specificity in individuals with calcification, common in advanced CKD	-Expensive -Limited availability -Device compatibility	-Invasive -Expensive
Safety in CKD	No issues	No issues	No issues	Iodinated contrast used	Gadolinium used	Iodinated contrast used; arterial access required

LV, Left ventricle; MPI, myocardial perfusion imaging.
Adapted from Dilsizian V, Gewirtz H, Marwick TH, et al. Cardiac imaging for coronary heart disease risk stratification in chronic kidney disease. *JACC Cardiovasc Imaging*. 2021;14:669–682.

symptoms.^{107,108} Troponin elevation is more common at lower eGFR values and is also associated with high LV mass index.¹⁰⁷ Notably, among individuals with advanced CKD, particularly among those receiving maintenance dialysis, elevated troponin levels in the absence of acute coronary syndrome are associated with a markedly increased risk of all-cause and cardiovascular disease mortality.¹⁰⁹ Kidney clearance contributes to cardiac troponin concentration, particularly at lower serum concentrations.¹¹⁰ Accordingly, the dynamic pattern of cardiac troponin concentration rather than presence of an elevated level, particularly if the initial elevation is not marked, is critical for diagnosing acute myocardial injury.¹¹¹

PREVENTION AND TREATMENT OF ISCHEMIC HEART DISEASE

Stages 3 to 4 chronic kidney disease

In the earlier stages of CKD, there is a moderate body of data, predominantly derived from subgroup analyses of larger clinical trials, demonstrating benefits with many interventions that are favorable in the general population. Therefore currently accepted strategies for primary and secondary prevention of cardiac disease in individuals with CKD stages 3 to 4 typically mirror those seen in the general

population, while exercising caution to minimize therapies with increased risk in patients with CKD.

In individuals with CKD stages 3 to 4 and ischemic heart disease, dyslipidemia (Table 42.4), hypertension, and diabetes likely should be treated similarly to current general population guidelines. On the basis of American Heart Association/American College of Cardiology joint guidelines, β -blockers are recommended for patients with chronic coronary artery disease with myocardial infarction in the past year or left ventricular ejection fraction $\leq 50\%$.¹¹² Either a calcium channel blocker or β -blocker is recommended as first-line antianginal therapy.¹¹² Specific indications for angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor blockers (ARBs) include ischemic cardiomyopathy, coexistent proteinuric kidney disease, and high-risk diabetes.¹¹³ Per the AHA/ACC guidelines, statins remain first-line therapy for lipid lowering in patients with coronary disease.¹¹² However, data from clinical trials suggest less efficacy of statins in persons with advanced CKD, specifically those treated with dialysis (see Table 42.4).¹¹⁴

A large trial of treatment with the glucagon-like peptide 1 receptor agonist (GLP-1RA), semaglutide, in persons with type 2 diabetes and CKD (eGFR 25–75 mL/min per 1.73 m²), all of whom had at least 100 mg/g of albuminuria at

Table 42.4 Randomized Controlled Trials of Statin Treatment Specifically in Chronic Kidney Disease

Study	Intervention	Population	Median Follow-Up	Primary Outcome	Risk of Primary Outcome	Risk of All-Cause Mortality
4D ⁴³	Atorvastatin 20 mg daily (vs. placebo)	Type 2 diabetes, HD <2 years, LDL 80–190 mg/dL	4.0 years	Composite of death from cardiac causes, fatal stroke, nonfatal MI, or nonfatal stroke	HR = 0.92 (0.77–1.10)	RR = 0.93 (0.79–1.08)
AURORA ¹¹⁷	Rosuvastatin 10 mg daily (vs. placebo)	HD or HDF >3 months	3.8 years	Composite of death from cardiovascular causes, nonfatal MI, or nonfatal stroke	HR = 0.96 (0.84–1.11)	HR = 0.96 (0.86–1.07)
ALERT ¹¹⁸	Fluvastatin 40 mg daily (vs. placebo)	>6 months post kidney transplant with stable graft function; no recent MI; total cholesterol 155–348 mg/dL	5.4 years	Major adverse cardiac event, defined as cardiac death, nonfatal MI, or coronary revascularization procedure	RR = 0.83 (0.64–1.06)	RR = 1.02 (0.81–1.30)
SHARP ¹¹⁹	Simvastatin 20 mg daily + ezetimibe 10 mg daily (vs. placebo)	No previous MI or coronary revascularization; Creatinine >1.7 mg/dL (men) or >1.5 mg/dL (women)	4.9 years	Composite of coronary death, nonfatal MI, ischemic stroke, or any revascularization procedure	RR = 0.83 (0.74–0.94)	RR = 1.02 (0.94–1.11)

Data in parentheses represents 95% confidence intervals. HR and RR report the relationship between treatment versus placebo, with values below 1 favoring treatment and above 1 favoring placebo. In SHARP, there was no statistically significant difference in the risk of the primary outcome between dialysis and nondialysis patients ($P = 0.25$), although, in subgroup analyses, there was no benefit in participants receiving HD at randomization (RR = 0.95 [0.78, 1.15]). 4D, German Diabetes Dialysis Study; ALERT, assessment of Lescol in renal transplantation; AURORA, a study to evaluate the use of rosuvastatin in subjects in regular hemodialysis: an assessment of survival and cardiovascular events; HD, hemodialysis; HDF, hemodiafiltration; HR, hazard ratio; LDL, low-density lipoprotein; MI, myocardial infarction; RR, risk ratio; SHARP, Study of Heart and Renal Protection.

Adapted from National Kidney Foundation. *Primer on Kidney Diseases*. 8th ed. Philadelphia: Elsevier; 2022.

enrollment, reported a significant reduction in the risk of the primary composite endpoint (major kidney disease events or death due to kidney-related or cardiovascular causes). Notably, semaglutide use was also associated with a reduction in the risk of death due to cardiovascular causes and in the risk of major cardiovascular events, highlighting a role for GLP-1RA treatment CV risk reduction among individuals with CKD and type 2 diabetes.¹¹⁵

Data regarding the efficacy of antithrombotic therapy are lacking in the CKD population but here again, particularly in CKD stage 3, likely mirror the general population, with aspirin at a dose of 75 to 100 mg daily likely first line among those with ischemic heart disease in sinus rhythm, and the use of dual antiplatelet therapy following coronary stenting.¹¹³ Critically, high-quality data balancing efficacy versus risk remain limited. This is important because individuals with advanced CKD are also at higher risk of bleeding complications. Limited studies indicate that antiplatelet agents appear to reduce the risk of myocardial infarction but do not reduce the risk of stroke or all-cause death.¹¹⁶

Individualized care and shared decision making are important given the challenges associated with therapies in individuals with CKD. For example, there is an increased risk of hyperkalemia with blockade of the renin-angiotensin-aldosterone system that needs to be balanced against the benefits of this therapy in the individual patient. Bleeding risks may outweigh ischemic heart disease benefits in some individuals. Hypotension and kidney perfusion may limit the ability to use multiple medications concurrently (such as ACE inhibitors or angiotensin receptor/neprilysin inhibitors, β -blockers, and diuretics), although each may have reasonable clinical and evidence-based data supporting their use.

Choosing between medical management and invasive management of coronary disease in advanced CKD remains challenging. The ISCHEMIA-CKD trial randomized 777 patients with moderate or severe ischemia on stress testing and CKD stage 4 to 5D including 373 with nondialysis CKD, to an initial invasive strategy consisting of coronary angiography and revascularization (if appropriate) added to medical therapy versus an initial conservative strategy

consisting of medical therapy alone, with angiography reserved for those whom medical therapy had failed.¹²⁰ The primary outcome, a composite of death or nonfatal myocardial infarction, was similar between groups. Critically, the invasive strategy was associated with a higher incidence of stroke and, among the subset not receiving dialysis at baseline, shorter time to the composite of all-cause death or dialysis initiation, suggesting an important role for conservative, noninvasive medical management for stable angina in the advanced CKD population. On the other hand, an initial invasive strategy may be preferable for persons with CKD with acute or unstable angina, unacceptable levels of angina that limit quality of life, left ventricular systolic dysfunction attributable to ischemia, or left main stem disease.

When angiography is clinically needed, the risk of contrast-induced acute kidney injury (AKI) should not be a reason to withhold it in most patients with CKD.¹²¹ When possible, attempts to minimize the risk of contrast nephropathy should be made through the avoidance of concurrent nephrotoxic agents, use of adequate volume administration before the administration of iodinated contrast agent, and minimization of the volume of contrast media. There is no benefit of bicarbonate or N-acetyl-L-cysteine over normal saline for prevention of AKI.¹²²

Stage 5 chronic kidney disease/dialysis

Little, if any, data exist showing that, among persons requiring dialysis, there is a survival benefit associated with cardiovascular disease therapies used in the general population.¹²³ The general failure to find specific interventions that significantly reduce the cardiovascular disease burden in individuals treated with maintenance dialysis probably reflects the numerous competing causes of death in these patients, such that addressing single risk factors may be insufficient to reduce mortality. Management of acute coronary syndrome is not specifically discussed here. For chronic risk factor management, statin therapy was evaluated in two large randomized clinical trials that enrolled only persons on hemodialysis (4D and AURORA), neither of which showed significant benefit.^{43,117} A third trial, SHARP, included dialysis ($n = 3023$) and nondialysis dependent ($n = 6247$) CKD. Overall, there was a 17% reduction in the relative risk of atherosclerotic cardiovascular events. Subgroup analysis appeared to show no benefit in those receiving dialysis at trial initiation, though the authors point out that statistical testing did not confirm any difference in the effects observed in dialysis versus nondialysis populations (see Table 42.4).¹¹⁹ Patients receiving peritoneal dialysis remain inadequately studied. On the basis of these results, KDIGO guidelines do not recommend routinely initiating statin therapy in patients undergoing hemodialysis, although the guideline suggests continuing statins in those who were receiving them before dialysis initiation.¹¹⁴ In patients with longer life expectancy, such as those expected to receive a kidney transplant, we suggest individualized decision making. Of note, KDIGO recommends statin therapy in transplant recipients.

Blood pressure targets and treatment strategies to reduce cardiovascular risk remain undefined in the dialysis population, with no high-quality data to date to guide treatment decisions.¹²⁴ Similarly, as discussed in more detail

later, antithrombotic therapies remain insufficiently studied in the dialysis population, resulting in a lack of certainty on whom to treat and which agents to use, for both primary and secondary prevention.¹²³

As noted earlier, the ISCHEMIA-CKD trial demonstrated that, particularly for patients with CKD with moderate ischemia, an initial conservative, noninterventional strategy may be preferable for managing stable angina.¹²⁰ If revascularization is indicated, older data suggest that coronary artery bypass grafting may be superior while more recent data are more nuanced, even suggesting that percutaneous intervention with a drug-eluting stent may be superior to coronary artery bypass grafting.¹²⁵ There are unfortunately no clinical trials to guide these decisions. Importantly, and despite common perceptions, undergoing cardiac surgery is not a definitive indication to transition a patient from peritoneal dialysis to hemodialysis.¹²⁶

PERIPHERAL ARTERY DISEASE

EPIDEMIOLOGY

The prevalence of PAD increases as kidney function declines with the highest prevalence in the dialysis population (see Fig. 42.6). CKD is associated with higher risk of incident PAD¹²⁷ including more severe PAD, requiring amputation. Individuals with CKD are less likely to be treated with revascularization, although those treated with revascularization have lower mortality than those not undergoing revascularization.¹²⁸ Patients with CKD and PAD also experience high rates of cardiovascular disease and mortality.¹²⁹

SCREENING

Ankle brachial index (ABI) is the most commonly used method to screen for PAD in the general population but, given the high prevalence of medial calcification particularly among individuals with advanced CKD, a high ABI in this population may reflect noncompressible vessels, with many individuals with PAD therefore not meeting traditional criteria for diagnosis of PAD (such as $ABI \leq 0.9$). Toe brachial index, which is not affected by medial calcification, could be considered an additional tool to assess for the presence of PAD in those with CKD.

PREVENTION AND TREATMENT

As in the general population, lifestyle modifications including exercise and smoking cessation should be encouraged.¹³⁰ Similarly, aggressive management of traditional cardiovascular risk factors such as diabetes and hypertension should be pursued, particularly before CKD stage 4 to 5. In both observational studies and trials of patients with all stages of CKD, statins have been shown to reduce the risk of PAD and may reduce amputation-free survival.^{119,131}

HEART FAILURE AND CARDIORENAL SYNDROME

EPIDEMIOLOGY AND MECHANISMS

Chronic kidney disease stages 3 to 4 and cardiorenal syndrome

Both incident and prevalent heart failure are common in people with CKD. In the Atherosclerosis Risk in Communities (ARIC) study, individuals with eGFR <60 mL/min

per 1.73 m² at baseline were at twice the risk of incident heart failure hospitalization and death compared with those with eGFR of ≥ 90 mL/min per 1.73 m², regardless of the presence of baseline coronary disease.¹³² Heart failure risk is higher among those with lower eGFR and with higher levels of albuminuria, rising dramatically at eGFR values below 60 mL/min/1.73 m².⁵

Heart failure and kidney disease are closely intertwined and, among patients with incident CKD, heart failure is highly prevalent.¹³³ It is unclear whether one organ dysfunction is causal to the other or whether heart failure and kidney disease occur in parallel. The term “cardiorenal syndrome” is used to define the occurrence of kidney disease and heart failure simultaneously and may include either acute or chronic changes in kidney and/or cardiac function. For example, cardiorenal syndrome is frequently diagnosed in the hospital setting when AKI occurs in parallel to acute decompensated heart failure. Up to 60% of patients admitted for acute decompensated heart failure have CKD, as defined by an eGFR of <60 mL/min per 1.73 m².¹³⁴ CKD, in turn, is one of the strongest risk factors for

mortality and cardiovascular events in individuals with acute decompensated heart failure.

Multiple mechanisms may contribute to both heart and kidney dysfunction including hemodynamic abnormalities, neurohormonal activation (including upregulation of the renin-angiotensin-aldosterone system), inflammation, endothelial dysfunction, and anemia (Fig. 42.9).¹³⁵ Hemodynamic insults are particularly important in the pathophysiology of cardiorenal syndrome. Certainly, low arterial pressures lead to renal vasoconstriction and hypoperfusion. However, it is now believed that venous congestion may be an important mediator of kidney dysfunction.¹³⁶

Observational studies suggest that venous congestion is associated with reduced eGFR in cross-sectional analyses.¹³⁶ Animal models have shown intrarenal hemodynamic changes with venous congestion including reductions in renal blood flow and increase in tubular sodium avidity. Venous congestion may stimulate upregulation of fibrotic pathways in the kidney and also activate endothelial cells to release local neurohormones and cytokines, which will propagate inflammatory pathways.¹³⁷

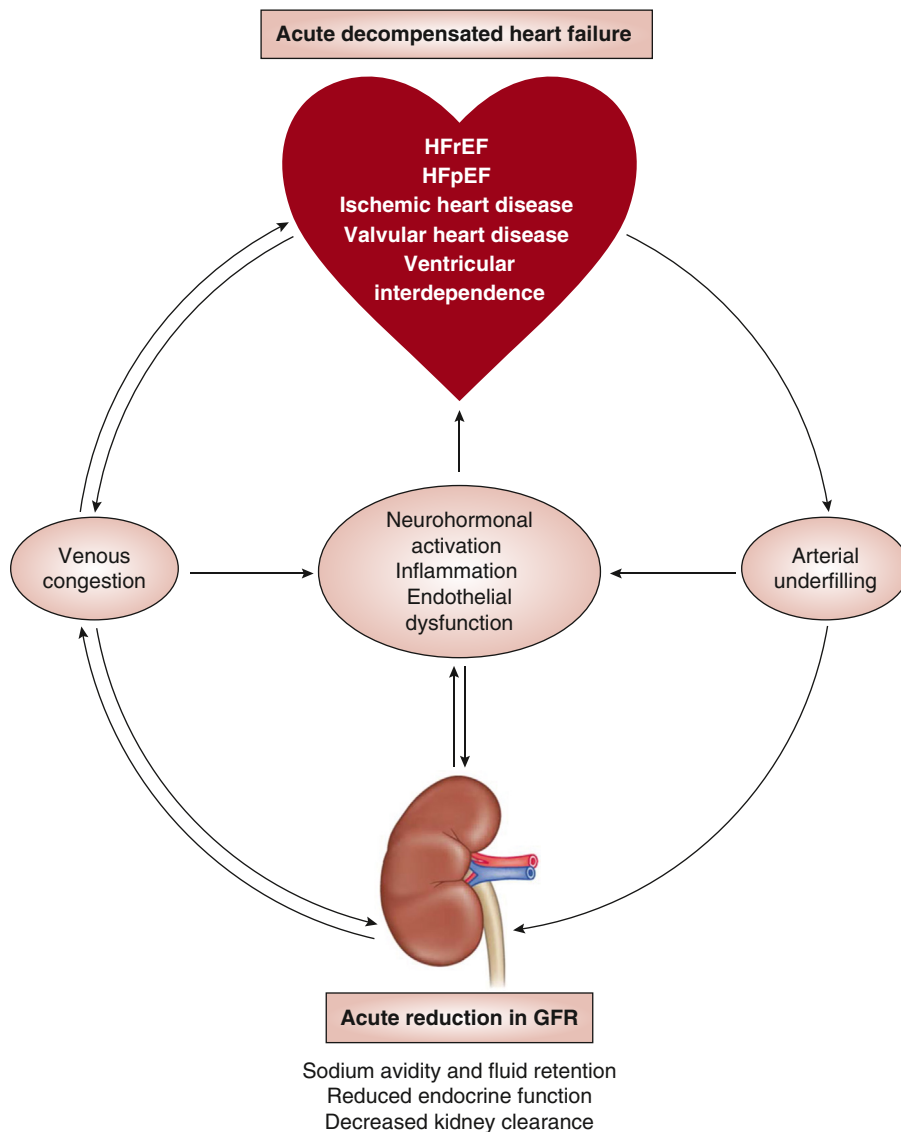


Fig. 42.9 Proposed pathways leading to the cardiorenal syndrome and its complications. The inciting event is usually an acute decompensation of heart failure, resulting in venous congestion and/or arterial underfilling that promotes neurohormonal activation, inflammation, and endothelial dysfunction. Clinically, this manifests with reductions in GFR, as well as increasing sodium avidity and fluid retention, all of which further perpetuate the process. (Reprinted from McCallum W, Sarnak MJ. Cardiorenal syndrome in the hospital. *CJASN*. 2023;18:933–945.)

Right-sided heart failure is also common in patients with kidney disease, most often resulting from either left-sided heart failure or pulmonary hypertension, which in turn may be due to left-sided heart failure, high cardiac output, and/or increases in pulmonary vascular resistance. Several studies in CKD suggest that pulmonary hypertension by echocardiographic criteria is common in this population. Among nearly 3000 CRIC participants with CKD, 21% had evidence of pulmonary hypertension (ePASP >35 mm Hg or tricuspid regurgitant velocity >2.5 m/s [a more inclusive definition than the more commonly used 2.8 m/s threshold], estimated from Doppler echocardiogram).¹³⁸ The prevalence of pulmonary hypertension increased with CKD severity, from 21% in CKD G3a to 32.8% in CKD G5.

Chronic kidney disease stage 5/dialysis

The incidence and prevalence of heart failure are also extremely high among patients undergoing dialysis. Studies estimate that a heart failure diagnosis is 12 to 36 times more common in patients receiving dialysis compared with the general population.¹³⁹ Based on U.S. Renal Data System (USRDS) administrative data that rely on billing codes to identify heart failure events, the 2-year cumulative probability of developing heart failure for patients initiating dialysis in 2016 was 50% for patients receiving hemodialysis and 36% for patients receiving peritoneal dialysis.⁹⁸

In addition to the impact of preexisting cardiovascular disease, there is evidence that hemodialysis itself may provoke a transient reduction in myocardial perfusion and contractility (myocardial stunning) even in the absence of coronary artery disease. The extent of myocardial stunning was related to the ultrafiltration volume and the risk increased markedly above an ultrafiltration volume of 2 L. Critically, the detection of intradialytic myocardial stunning was associated with a permanent decrease in left ventricular ejection fraction and a higher mortality rate over 12 months.⁵¹

DIAGNOSIS OF HEART FAILURE IN CHRONIC KIDNEY DISEASE

Despite its high prevalence, there are significant challenges to HF diagnosis among individuals with CKD, potentially leading to underdiagnosis (and thus undertreatment). HF is a clinical diagnosis based on a characteristic history and physical examination indicative of inadequate cardiac output or cardiac filling to meet systemic demands. Accurate diagnosis relies on signs or symptoms of volume overload, such as shortness of breath, orthopnea, paroxysmal nocturnal dyspnea, and edema, and signs of elevated cardiac filling pressures, such as jugular venous distention, and displaced apical impulse. These symptoms and signs can also be seen in individuals with CKD, particularly in late-stage CKD and dialysis, when it can be difficult to distinguish HF from systemic volume overload.

Early HF symptoms in particular are often missed or underappreciated in patients with CKD. In a well-defined CKD population of nearly 3,000 patients without a clinical diagnosis of HF, 25% reported significant HF symptoms as determined by the validated 23-item Kansas City Cardiomyopathy Questionnaire (KCCQ).¹⁴⁰ Furthermore, this study showed an inverse association between eGFR level and the burden of HF symptoms. In a follow-up study,

greater KCCQ symptom burden was associated with a higher short-term risk of incident HF (odds ratio of 3.30 for those with the highest symptom burden compared with those with the lowest symptom burden).¹⁴⁰ These data suggest that the KCCQ may detect subclinical HF with symptoms that would otherwise be attributed to kidney dysfunction and shows promise for future HF screening in CKD.

Ancillary imaging and laboratory tests such as echocardiograms and N-terminal pro-BNP (NT pro-BNP), which are helpful to confirm the diagnosis of HF in the general population, are often difficult to interpret in CKD, particularly among patients receiving dialysis. Given the high prevalence of LVH, it is unclear whether these biomarkers can be used reliably to discriminate risk of subsequent HF. Circulating concentrations of NT pro-BNP are affected by lower kidney excretion. Despite this concern, a number of studies have shown that elevations in NT pro-BNP are in fact strongly associated with incident HF in patients with CKD, even after accounting for eGFR level and urine ACR. However, the use of these NT-proBNPs in the clinical setting still remains controversial since there are no accepted cutoffs for identifying HF in the CKD population and no trials using these levels to guide management.¹⁰⁸

HEART FAILURE AND KIDNEY FUNCTION DECLINE

Heart failure itself contributes to the development and progression of CKD, highlighting a bidirectional interaction between the heart and the kidneys. The prevalence of CKD among those with chronic HF is estimated to be 40%.¹⁴¹ Studies have shown that worsening HF is associated with progressive loss of kidney function. In the CRIC study of individuals with CKD, a history of HF was independently associated with a 29% higher risk of a 50% decline in eGFR or initiation of dialysis compared with those without HF.¹⁴² There are several postulated mechanisms to explain this bidirectional association between kidney disease and HF. For example, hemodynamic changes are common in patients with HF and can lead to increased kidney venous congestion and poor forward flow, which may compromise kidney function. Additionally, adverse effects from HF pharmacotherapies, like aggressive diuresis, may cause AKI and subsequent progression of CKD. Similarly, kidney disease leads to increased sodium retention and higher blood pressure and afterload, which can contribute to cardiac damage. Several other hormonal and inflammatory pathways may be activated, leading to both kidney and cardiac stress and damage.

In addition to clinical HF, biomarkers and imaging suggestive of subclinical HF are also associated with a progressive decline in kidney function. In the Cardiovascular Health Study of older adults, those in the highest quartile of NT pro-BNP had a 38% higher risk of incident CKD and a 67% higher risk of rapid eGFR decline compared with those in the lowest quartile.¹⁴³ Another study of diverse participants with no CVD reported that those in the highest quartile of cardiac concentricity as assessed by cardiac MRI had an additional 21% decline in kidney function compared with those in the lowest quartile.¹⁴⁴ Similarly, LVH measured with echocardiograms among African-Americans in the Jackson Heart Study was associated with kidney function decline.¹⁴⁵ Thus developing more effective HF therapies may also help

preserve kidney function, which in turn would lead to better overall prognosis.

PREVENTION AND TREATMENT OF HEART FAILURE IN NONDIALYSIS CHRONIC KIDNEY DISEASE

Potentially modifiable risk factors for heart failure in patients with nondialysis requiring CKD include hypertension, extracellular volume overload, abnormal mineral metabolism including hyperphosphatemia and secondary hyperparathyroidism, and anemia.

Blood pressure

Blood pressure—lowering in general may reduce the risk of heart failure in patients with CKD. Data from the large SPRINT trial, which enrolled persons at increased risk of cardiovascular disease but without diabetes, including a subgroup with CKD (eGFR >20 mL/min/1.73 m²), demonstrated a significant reduction in the primary outcome of major cardiovascular events, as well as a reduction in risk of heart failure (HR 0.62, 95% CI 0.45–0.84) when participants were randomized to a systolic blood pressure target of <120 mm Hg versus a target of <140 mm Hg.¹⁴⁶ In patients requiring maintenance dialysis, data on blood pressure targets to reduce heart failure risk remain uncertain given the lack of definitive clinical trial data.

RAAS blockade and neprilysin inhibitors

Specific agents may be indicated for the treatment of heart failure including in individuals with CKD. As in the general population, RAAS inhibition remains central for both kidney and cardiovascular protection. ACE inhibitors and ARBs may have cardiac benefits independent of their blood-pressure-lowering effects in persons with systolic heart failure and CKD stages 1 to early stage 4. Potential further benefits may be associated with mineralocorticoid receptor blockade including with use of spironolactone, eplerenone, and finerenone; however, hyperkalemia may limit use of these agents, particularly in conjunction with ACE inhibitors or ARBs.¹⁴⁷ Notably, recent data evaluating finerenone use in individuals with diabetic kidney disease showed a significant reduction in the composite of death, kidney failure, or a 40% reduction in GFR, as well as in a secondary composite outcome that also included hospitalization for heart failure.¹⁴⁸

Sacubitril/valsartan, a combination agent consisting of a neprilysin inhibitor and an ARB, is a guideline-directed medical therapy for HFrEF, considered preferable to an ACE inhibitor or an ARB alone in the absence of contraindications. Although there are only limited data in individuals with CKD, the United Kingdom Heart and Renal Protection III trial, a randomized double-blind trial in 414 participants with mean estimated GFR of 35 mL/min/1.73 m² without HF, suggested that sacubitril/valsartan had similar safety in this population as irbesartan monotherapy.¹⁴⁹ There are no clinical trial data to date exploring use of these agents in individuals with advanced CKD and HFrEF.

β-blockers

β-blockade is another mainstay of heart-failure therapy in the general population. These agents are also likely beneficial in patients with nondialysis CKD,¹⁵⁰ while evidence from one small trial supports carvedilol use to reduce mortality risk

in patients with heart failure with reduced ejection fraction receiving hemodialysis.¹⁵¹

SGLT2 inhibitors

Initially developed as agents for improving glycemic control in persons with diabetes, SGLT2 inhibitors have shown substantial cardiovascular disease benefits that are largely driven by reductions in heart failure and cardiovascular death. The SGLT2 inhibitors block sodium/glucose uptake in the proximal tubule, inducing plasma volume contraction and restoring tubuloglomerular feedback to decrease glomerular hyperfiltration. Moreover, SGLT2 inhibitors may improve the efficiency of myocardial energetics,¹⁵² which may in part explain their beneficial effects on cardiac outcomes. Several CKD-focused SGLT2 inhibitor trials have recruited patients with CKD with and without diabetes at eGFR levels as low as 20 mL/min/1.73 m², with the majority of participants exhibiting at least moderately elevated albuminuria. In addition to slowing the progression of CKD, SGLT2i have demonstrated consistent and strong reductions in risk of HF hospitalizations and cardiovascular mortality,^{153–155} despite only modest effects on long-term glycemic control in individuals with diabetes.¹⁵⁶ Similarly, SGLT2 inhibitor trials in individuals with heart failure have shown a significant reduction in adverse kidney outcomes.^{157,158} On the basis of these favorable trial results, SGLT2 inhibitors are recommended as first-line agents in individuals with kidney disease and heart failure. Data also suggest a benefit of SGLT2i in patients with HFpEF (the most common HF type seen in kidney disease), although few participants had advanced CKD.¹⁵⁹

Mineral metabolism and anemia

Although abnormal mineral metabolism and anemia are thought to contribute to the risk of heart failure, randomized trials in kidney disease patients targeting these conditions have not demonstrated a clear treatment benefit. Trials in patients with nondialysis CKD and patients receiving dialysis did not find a benefit of vitamin D repletion or treatment of secondary hyperparathyroidism on LVH or other cardiovascular endpoints, including trials assessing paricalcitol and cinacalcet^{42,160,161} (see [Chapter 52](#) for further discussion). In patients with CKD, targeting normalization of hemoglobin concentrations with recombinant human erythropoietin had no effect on the similar surrogate outcome of LVH or left ventricular mass¹⁶² (see [Chapter 53](#) for further discussion). Critically, no trials in CKD stages 3 to 4 have demonstrated a reduction in cardiac outcomes or mortality with these interventions when they are used for the purpose of treating or preventing LVH.

TREATMENT OF ACUTE DECOMPENSATED HEART FAILURE WITH CONCURRENT ACUTE KIDNEY INJURY

Treatment of patients hospitalized with heart failure, particularly in the setting of AKI or CKD, requires special attention. Decongestion is the cornerstone of treatment, and patients who achieve adequate decongestion have better in-hospital and long-term outcomes.¹⁶³ Intense tubular sodium avidity requires judicious use of diuretics and combining agents with varying pharmacologic targets. The first step is usually intravenous loop diuretics, which

block the Na-K-2Cl channel. If sufficient decongestion is not achieved with loop diuretics alone, a “sequential nephron” blockage strategy can be effective with addition of thiazide diuretics and acetazolamide. For example, a large clinical trial demonstrated greater natriuresis in patients treated with acetazolamide in addition to furosemide versus furosemide alone.¹⁶⁴ SGLT2 inhibitors can also enhance natriuresis by blocking proximal tubular sodium reabsorption (and have other kidney and myocardial benefits as discussed earlier). Agents that target the neurohormonal system, such as spironolactone, neprilysin inhibitors (which also augment the endogenous natriuretic peptide system) and tolvaptan, are also possible therapies; however, data regarding the efficacy of these agents remain mixed, particularly in the setting of very low GFR. While extracorporeal ultrafiltration may be an option in refractory volume overload, the CARESS trial demonstrated that early ultrafiltration (vs. diuretics) was not more effective in decongestion and was associated with slightly increased serum creatinine levels.¹⁶⁵

TREATMENT OF HEART FAILURE IN PATIENTS RECEIVING DIALYSIS

Although there are limited clinical trial data, HF therapies such as RAASi and β -blockers should generally be used in patients receiving dialysis unless there are contraindications (e.g., severe hypotension). To date, the safety and efficacy of SGLT2 inhibitors are in this population is unknown. A limited number of studies have explored the impact of correcting novel HF risk factors in dialysis like anemia, acid-base disturbances, and bone metabolism abnormalities. One randomized controlled trial evaluating cinacalcet in reducing parathyroid hormone abnormalities in ESRD failed to reduce the risk of HF.¹⁶⁶ As of February 2025, there are no published trial data on the use of ARB/neprilysin inhibitor (ARNi) therapy among patients receiving dialysis.

Once patients with HF reach kidney failure, the mode and frequency of dialysis may be important to mitigate HF risk. Observational studies, albeit with many limitations, suggest that peritoneal dialysis and more frequent and extended dialysis may be beneficial under certain circumstances.¹⁶⁷ Several studies suggest that peritoneal dialysis in HF may significantly improve New York Heart Association Class and body weight and reduce the number of days hospitalized for HF.¹⁶⁸⁻¹⁷⁰ Hypothesized reasons for these improvements include better volume control with continuous gentle ultrafiltration that minimizes rapid fluctuations in volume and improved control of potassium and sodium hemostasis.

The impact of hemodialysis frequency on HF risk has also been studied. A 2010 randomized clinical trial compared hemodialysis six times per week versus three times per week and found that more frequent dialysis was associated with a composite outcome of less increase in LVMI and lower risk of death.¹⁷¹ Furthermore, in an observational study more frequent hemodialysis and home hemodialysis were associated with lower ultrafiltration volumes and home hemodialysis was associated with a lower incidence of myocardial stunning.¹⁷² Thus the use of more frequent hemodialysis and avoidance of high ultrafiltration volumes may help to preserve cardiac function in persons requiring maintenance hemodialysis.

ARRHYTHMIAS AND SUDDEN CARDIAC DEATH

EPIDEMIOLOGY

Arrhythmias are extremely common in individuals with CKD, likely reflecting a high prevalence of structural heart disease, ischemic heart disease, and electrolyte abnormalities. Atrial fibrillation is the most common arrhythmia,¹⁷³ with rising prevalence as GFR declines.¹⁷⁴ Prevalence estimates for paroxysmal and permanent atrial fibrillation are as high as 30% in individuals with advanced CKD including patients undergoing dialysis.¹⁷⁵⁻¹⁷⁸ Rates vary based on source, with administrative data based on billing codes reporting prevalence rates of approximately 10%, likely reflecting limitations of this data source.¹⁷⁹ In contrast, the Monitoring in Dialysis Study, in which 66 patients treated with hemodialysis received an implantable loop recorder, detected atrial fibrillation in 41% of participants.¹⁸⁰ Atrial fibrillation is associated with a significant increased risk of death and other types of cardiovascular disease (including but not limited to strokes) in individuals with advanced CKD.^{179,181}

Bradycardia, asystole, and ventricular arrhythmias are also common and are a leading cause of death in individuals with advanced CKD, particularly those receiving dialysis.¹⁸² Notably, the Monitoring in Dialysis Study of patients treated with hemodialysis described bradycardia as the most common clinically significant arrhythmia and noted that arrhythmia was most common at the end of the long interdialytic interval and during the first hemodialysis session of the week.¹⁸⁰ On the basis of 2020 USRDS data, cardiac arrest and arrhythmia account for 33% of all deaths (and ~42% of deaths with a known cause) among patients receiving dialysis, highlighting arrhythmia as a critical issue for this population, even in a year when COVID-19 was a substantial cause of death.¹⁸²

PREVENTION AND TREATMENT

Ventricular arrhythmia and sudden cardiac death

There are few data on prevention and treatment of arrhythmias and sudden cardiac death in the CKD or dialysis population, with most current treatment recommendations for individuals not treated with dialysis mirroring those seen in the general population. Among patients receiving hemodialysis, the temporal relationship of arrhythmias to the dialysis cycle suggests that volume status and electrolytes, as well as rapid fluctuation of electrolyte levels, may be modifiable factors to reduce the risk of arrhythmias and sudden cardiac death. Although an increasing number of patients with late-stage CKD and patients undergoing dialysis with reduced ejection fraction heart failure are receiving implantable cardioverter-defibrillators (ICDs) to prevent sudden cardiac death, there are no trial data that have shown a survival benefit or demonstrated cost-effectiveness. Observational data suggest no survival benefit associated with ICD placement in individuals with HFrEF and CKD.¹⁸³ Critically, ICD and other cardiac device wires typically traverse the left subclavian vein and may predispose to central venous stenosis, adversely affecting hemodialysis vascular access options. Newer, leadless devices

may be particularly good options for the advanced CKD and dialysis populations, where vein preservation is paramount. Given the high incidence of sudden cardiac death, one key preventative strategy for ambulatory settings where patients with CKD are treated, including clinics and dialysis facilities, is to ensure the availability of an automated external defibrillator and trained clinic personnel.

Atrial fibrillation

There are also limited data on the efficacy and safety of treatment of atrial fibrillation in patients with kidney disease. Some data suggest that standard therapies for atrial fibrillation are underprescribed in persons with CKD.¹⁸⁴ Atrial fibrillation management focuses on rate control and/or rhythm control, as well as stroke prevention. Treatment decisions likely mirror those seen in the general population, with the recognition that rate or rhythm-control agents that lower blood pressure may be poorly tolerated among patients receiving dialysis and appreciating the variable dialysis clearance of specific β -blockers that may require attention to dosing strategies. The current literature related to stroke risk and prevention including anticoagulation is summarized in the next section discussing stroke in people with CKD. Unfortunately, despite the high prevalence of atrial fibrillation in individuals with advanced CKD, there is a paucity of high-quality data to provide evidence-based treatment recommendations for this population.¹⁸⁵

STROKE

Epidemiology

Cerebrovascular disease is common in individuals with CKD (see Fig. 42.8), with a higher incidence of both ischemic and hemorrhagic events than seen in the general population. Critically, even in the absence of clinically evident strokes, both silent lesions and substantial brain white matter disease may be present. Not surprisingly, the presence of cardiovascular disease in individuals with CKD is associated with cerebrovascular manifestations including worse cognitive function.

Prevention and treatment

Although not specifically studied, stroke prevention and treatment strategies for patients with earlier stages of CKD likely should follow general population guidelines including management of traditional risk factors and the use of antithrombotic agents as indicated.¹⁸⁶ Major evidence gaps exist in patients with advanced CKD including those receiving dialysis: 1. Does anticoagulation for primary prevention improve outcomes? 2. What is the optimal agent for anticoagulation? Data to support decision making in this population are limited, with poor predictive performance of commonly used stroke risk prediction instruments including the CHA₂DS-VASc score, potentially reflecting the high competing risk of death not related to stroke.¹⁸⁷⁻¹⁸⁹ Similarly, currently used scores estimating bleeding risk in individuals with CKD likely underestimate risk substantially, particularly among those with advanced kidney disease or those receiving dialysis.¹⁸⁹

In dialysis cohorts, data on the value of warfarin for primary thromboembolism prevention in atrial fibrillation are mixed, with some (but not all) observational data suggesting an increased risk of death in patients treated with warfarin compared with no anticoagulation agent.^{190,191} If true, this may reflect not only an increased risk of bleeding events but also the relationship

between warfarin and increased vascular calcification, mediated by inhibition of vitamin K-dependent carboxylation of matrix Gla protein, a calcification inhibitor. However, the observational studies on the benefits and safety of warfarin in this population have yielded conflicting results, leading to uncertainty on use of anticoagulation for prevention of stroke.

Definitive clinical trial data examining the use of either warfarin or DOACs among patients undergoing dialysis are lacking, with the few that have been conducted being underpowered and therefore inconclusive.¹⁹² Critically, the competing risks of bleeding complications and all-cause mortality due to other risk factors are high in the advanced CKD and dialysis population, contributing to uncertainty regarding the optimal treatment strategies.¹⁹²⁻¹⁹⁴ Limited safety and efficacy data exist for DOACs in CKD stage 4 to 5 or dialysis. Apixaban has been approved by the U.S. Food and Drug Administration for use in dialysis, albeit in the absence of safety data from clinical trials. The current AHA/ACC guideline highlights the ongoing equipoise, stating that: “For patients with AF at elevated risk for stroke and who have end-stage CKD or are on dialysis, it might be reasonable to prescribe warfarin (INR 2.0–3.0) or an evidence-based dose of apixaban for oral anticoagulation to reduce the risk of stroke.”¹⁸⁶ Despite this lack of data, apixaban treatment has been widely adopted in the dialysis population, though uncertainty remains regarding whether the appropriate dose is 2.5 mg or 5 mg twice daily, with differing recommendations in the United States and Europe.^{192,193,195} There are even fewer data on patients receiving peritoneal dialysis, with one small study suggesting that reduced dosing of apixaban and frequent monitoring may be necessary.¹⁹⁶

Left atrial appendage occlusion has emerged as a mechanical alternative to systemic anticoagulation for preventing atrial fibrillation-associated thromboembolism. Results from several small studies suggest that this procedure is safe and diminishes the need for long-term systemic anticoagulation, although anticoagulation is used for 1 to 2 months following the procedure, followed by 6 months of dual antiplatelet therapy before transition to aspirin monotherapy. More data are needed on the efficacy and safety of left atrial appendage occlusion in patients with CKD including those receiving dialysis.¹⁹⁷

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QUESTIONS

1. A 54-year-old man with a history of chronic kidney disease secondary to polycystic kidney disease is seeing you for a follow-up visit. His estimated glomerular filtration rate is stable at 42 mL/min/1.73m² with minimal proteinuria. His blood pressure is 128/68 mm Hg on two antihypertensive medications. He is concerned about his future cardiovascular risk. What additional testing should be performed?
 - a. Check lipid profile
 - b. Electrocardiogram
 - c. Echocardiogram
 - d. 24-hour ambulatory blood pressure monitor

Answer: a

Rationale: Hyperlipidemia is an important cardiovascular risk factor in persons with kidney disease. In individuals with CKD stages 3 to 4, data from clinical trials suggest that dyslipidemia should be treated similarly to current general population guidelines. Answers B and C are incorrect as there are no studies to support screening electrocardiograms or echocardiograms in asymptomatic CKD patients. Answer D is incorrect since his blood pressure has been stable and well controlled.

2. A 67-year-old woman with a history of diabetic kidney disease presents with new dyspnea on exertion. She denies other symptoms or changes in health. Her eGFR is stable at 45 mL/min/1.73 m². She has been taking an ACEi, SGLT2i, and insulin as prescribed. What is the next step in evaluation?
 - a. Echocardiogram
 - b. Nuclear stress test
 - c. Coronary artery CT scan
 - d. Cardiac catheterization

Answer: b

Rationale: This patient should undergo further evaluation for coronary artery disease. Nuclear stress tests are widely available and have high sensitivity and specificity in persons with CKD. An echocardiogram may be indicated if there is suggestion of heart failure on the stress test. The use of coronary artery CT scans to diagnose coronary artery disease is uncertain in patients with CKD. Cardiac catheterization may be indicated; however, noninvasive testing should be pursued first.

3. A 60-year-old woman with a history of heart failure with reduced ejection fraction (EF 20%) has progressed to kidney failure and is planning for kidney replacement therapy. What kidney replacement modality would you suggest?
 - a. In-center hemodialysis thrice weekly
 - b. Conservative kidney management
 - c. Peritoneal dialysis
 - d. Preemptive kidney transplant

Answer: c

Rationale: Given her low ejection fraction, ultrafiltration may be challenging with in-center thrice-weekly hemodialysis. Studies have shown better outcomes when more frequent dialysis modalities, such as peritoneal dialysis, are utilized in patients with heart failure. Conservative kidney management may be an option, but this would need to be

discussed with the patient to assess her goals. The patient is unlikely to be a candidate for preemptive kidney transplant due to her heart failure.

4. What are the best strategies to reduce risk of sudden cardiac death among in-center hemodialysis patients?
 - a. Early implantable cardioverter defibrillator placement
 - b. Initiation of amiodarone
 - c. Annual screening electrocardiograms
 - d. Availability and training on use of automated external defibrillator in the dialysis unit

Answer: d

Rationale: Studies have not shown clear benefit for reduction of sudden cardiac death with implantable cardioverter defibrillator placement in dialysis patients. There is no role for prophylactic use of amiodarone. Screening electrocardiograms are not recommended in the absence of symptoms.

5. A 62-year-old man with CKD stage G5D receiving maintenance hemodialysis presented to the emergency department with abdominal pain that resolved shortly after arrival. Blood pressure is 142/74 mm Hg, and his examination is unremarkable. His chest radiograph suggests mild pulmonary vascular congestion, and he is due for hemodialysis the following day. Abdominal ultrasound shows several gallstones and biliary sludge. Electrolytes and liver function tests are within normal limits, and hemoglobin is 11.1 g/dL. The initial troponin (high sensitivity Troponin T, upper limit of normal of 12 ng/L) is 74 ng/L. Subsequent troponin T levels 3 and 8 hours later using the same assay are 71 ng/L and 76 ng/L. An electrocardiogram shows lateral T wave inversions, unchanged from prior electrocardiograms, and findings consistent with left ventricular hypertrophy. These clinical results suggest:
 - a. Acute non-ST elevation myocardial infarction
 - b. Normal heart perfusion
 - c. Increased long-term risk of a subsequent cardiovascular disease event
 - d. a and c

Answer: C

Rationale: This patient is likely presenting with uncomplicated gallstone disease. To diagnose a myocardial infarction, elevated troponin levels need to be accompanied by a dynamic change in the troponin levels over time. In this case, the troponin levels were relatively stable over an 8-hour period. While kidney clearance does contribute to troponin levels and levels of troponin in serum are often elevated in asymptomatic individuals tested with maintenance dialysis, several studies have shown that elevated troponin levels are strong predictors of subsequent adverse cardiovascular disease events. This modest elevation in troponin may reflect low-grade increased production of troponin due to chronic demand-perfusion mismatches, as can be seen with increased left ventricular mass and underlying coronary artery disease including small vessel disease. There may also be functional factors affecting cardiac myocyte perfusion including endothelial dysfunction.